

Default Zero-Shot LLM APIs Over-Flag Indian Financial SMS: A Policy-Aligned Benchmark and Template-Level Pre-Deployment Audit

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Abstract

Generic ham/spam labels do not define a financial-fraud blocking policy. We present a 1,393-message policy-aligned Indian SMS benchmark in Hindi, Hinglish, and English-India, separating source labels, fraud/non-fraud/unclear adjudication, and allow/review/block actions. GPT-4o and Gemini helped adjudicate source-positive messages, so positive metrics are secondary. Our primary endpoint is false-positive rate (FPR) on 1,294 policy-negatives: 1,222 untouched source negatives plus 72 source positives converted to safe. Under a common minimal prompt, six LLM APIs exceed illustrative 1% FPR gates on Bank and UPI templates. GPT-4o-mini records 77.3% and 56.9% FPR; a fine-tuned 270M XLM-R records 0.0% on both, 0.4% globally, and 0.980 recall. Policy definitions and seven examples reduce GPT-4o to 0.0% Bank/UPI and 1.5% global FPR at 0.949 recall; Claude reaches 1.8% at 0.960 recall. Prompting changes which gate fails. GPT-4o P3 and the AND consensus meet the FPR gates at lower policy-label recall; the local model meets both FPR and recall gates on this benchmark. Because positive labels partly use GPT-4o/Gemini, recall and F1 are secondary, not an overall ranking. We release labels, prompts, parsed decisions, analysis code, and checkpoints without a production or unseen-institution claim.

1 Introduction

SMS-based financial fraud in India is a live operational problem. Government figures report 1.342 million UPI fraud incidents totaling INR 10.87 billion in FY2023–24 (Govt. of India, 2024); the Reserve Bank of India’s January 2025 circular calls for safeguards on voice and SMS channels (RBI, 2025); and India’s Central Bureau of Investigation reported dismantling a “Phishing SMS Factory” using 21,000 illegal SIM cards in December 2025 (CBI, 2025). These sources

motivate the audit setting; we do not conduct a regulatory-compliance or production study.

A natural deployment instinct is to filter SMS at the channel layer using a commercial LLM API: ask whether each message is a scam, without collecting local labels or tuning a prompt. This is operationally attractive—no model training or hosted inference stack—but it leaves the provider to infer the operator’s fraud policy. We ask a bounded question: do the tested APIs, under a common default zero-shot prompt and stronger policy-grounded prompts, pass conservative release gates on this benchmark?

What we mean by deployment audit. A deployment audit measures whether a candidate filter satisfies reference gates on a representative held-out set before production traffic touches it. For financial SMS, aggregate F1 can hide concentrated customer harm. We use (i) per-template FPR on legitimate institutional messages, (ii) global FPR, and (iii) fraud recall; projected manual-review volume is a reported operational quantity, not a pass/fail gate. The 1% FPR and 0.95 recall thresholds are illustrative conservative references, not regulatory requirements or universal standards; operators must set them from their traffic and harm costs.

Three constructs, not one label. The source corpora provide generic *ham/spam* annotations. We map their positive class to a study-specific *fraud/non-fraud/unclear* adjudication policy (the historical pipeline stores the non-fraud bin as “marketing”), then evaluate binary fraud scores that a deployment may translate into *allow/review/block* actions. Calling a marketing-labeled source item “safe” under this policy does not assert that the source label was erroneous or that the message is desirable; it means the item does not meet our financial-fraud blocking definition.

Headline audit finding. Under the original minimal-prompt condition, all six APIs exceed

both populated institutional gates. GPT-4o-mini records 77.3% Bank/account FPR ($n = 163$) and 56.9% UPI/Payment FPR ($n = 146$); the local XLM-R records 0.0% on both. A post-review prompt ladder changes this result sharply: policy definitions plus seven examples reduce GPT-4o’s Bank/UPI FPR to 0.0%, although its global FPR is 1.5% and recall is 0.949. Other models fail different gates. The scientifically relevant result is therefore prompt- and policy-sensitivity, not a universal local-versus-LLM ranking.

Contributions. (1) A policy-aligned multilingual benchmark and transparent conversion of generic spam labels into fraud/non-fraud/unclear decisions (§3.1–§3.2). (2) A deterministic institutional-template taxonomy that makes per-template FPR a release gate rather than an error-analysis footnote (§3.3). (3) A same-set comparison of default zero-shot APIs, stronger policy-grounded prompts, dual-API consensus, and locally supervised XLM-R (§4). (4) Auditable operating-point, annotation, negative-label, structural-duplicate, and calibration analyses. We release frozen labels, prompts, parsed predictions, aggregate-analysis scripts, and checkpoints; controlled-source re-inference is not self-contained. We make no production or unseen-institution claim.

2 Related Work

Local fine-tuned models versus zero-shot LLMs. Bucher and Martini (2024) show that fine-tuned sub-1B encoders significantly outperform zero-shot generative models on text classification; Ma et al. (2025) study conventional and LLM baselines for psychological-scam classification in English; Belcak et al. (2025) argue that small language models are economical for narrow, repetitive agentic tasks. We extend this line into multilingual in-market Indian financial SMS by quantifying template-specific false positives and showing how policy-grounded prompting changes them.

Multilingual phishing/smishing. Trad and Chehab (2024) compare prompt engineering and fine-tuning for phishing-URL detection. Harichandana et al. (2024) present a compact on-device SMS and URL smishing pipeline using a 3.46 MB classifier; Goenka et al. (2024) provide a broader phishing taxonomy. To our knowledge, no prior work releases a policy-aligned multilingual Indian SMS benchmark separating

fraud adjudication from generic spam labels, or measures LLM API false-positive rates by Indian institutional template.

Adversarial robustness (out of scope). Morris et al. (2020); Jin et al. (2020) establish standard black-box attacks against transformer classifiers, and Zhan et al. (2025) demonstrate that adaptive attacks routinely break static defenses on LLM-agent indirect prompt injection. We do not run adaptive attacks (§6); adversarial evaluation in the multilingual SMS setting is future work.

Standards and regulatory anchors. The Reserve Bank of India’s January 2025 circular (RBI, 2025), NIST AI RMF (Tabassi, 2023), and MITRE ATT&CK T1566 (MITRE, 2026) motivate risk management and threat modeling. We use them as context, not as evidence of legal compliance.

3 Method

3.1 Data sources and per-source label policy

We assemble training data from four public Indian SMS corpora plus a 30K English subsample drawn from a global scam-detection pool. Both components enter the policy-aligned multilingual training mix (§3.4); the English subsample is included for cross-lingual transfer coverage during fine-tuning. The English-only XLM-R control reported in §4 is a separate model trained on the English subsample alone and serves as the ablation that isolates the contribution of in-market Indian SMS data. The four Indian sources use generic “ham”/“spam” or 0/1 labels that do not distinguish marketing from fraud. Table 1 reports the per-source raw positive count, the final adjudication-pipeline breakdown after policy conversion (§3.2), and the resulting assignments.

The most striking row is *iiitd*: under our study policy, the full adjudication pipeline assigns 5.7% of its 995 “spam” positives to fraud and 77.5% to the non-fraud bin, principally commercial messages from named senders (including Crocs, Aircel, Thyrocare, and Meru Cabs). This is a task redefinition from spam filtering to fraud blocking, not evidence that *iiitd*’s original spam annotations are wrong. Borderline or potentially deceptive promotions can be assigned “unclear” and are dropped. By contrast, 98.3% of *cloveai_india* positives map to fraud. Mapping both spam and smishing to the same downstream warning action, as in COPS (Harichandana et al., 2024), illustrates

Table 1: Policy conversion for source-positive Indian SMS. The original positive class in iitd, vinit9638, and vinit9638_multi is generic “spam” that mixes marketing with fraud; we relabel positives via dual-judge consensus (GPT-4o + Gemini 2.5 Flash; Claude held out as a non-judge comparator) with 18 manually identified non-fraud prefix patterns as overrides. The historical pipeline stores these overrides in its “marketing” bin although the set also includes legitimate institutional notices. “Fraud” enters training as 1, “Non-fraud” as 0, and “Unclear” is dropped. Only source positives are converted; source negatives retain their original label.

Source	Raw	Frd	Non-frd	Uncl
cloveai (en_in)	5,558	5,463	1	94
vinit9638_multi (hi)	625	dedup w/ vinit9638		
vinit9638 (hi)	626	324	21	281
iitd (hi_en)	995	57	771	167
flatmars (mixed)	55	52	1	2
Total	7,234	5,896	794	544
		81.5%	11.0%	7.5%

how a different action policy can materially change downstream conclusions.

After per-source policy conversion and normalized fingerprint deduplication, the combined pool yields 1,393 messages in the held-out test set (stratified by language and label with seed=42), with 99 positives (7.1% prevalence), down from 221/1,443 (15.3%) in the original test: 50 source-positive rows judged unclear were dropped and 72 were converted to safe. Because F1 is base-rate-sensitive at low prevalence, we report institutional FPR and MCC alongside F1 throughout §4. A stricter post-hoc structural clustering audit is reported in §4.3; it is a sensitivity analysis, not an unseen-template split.

3.2 Positive-class policy conversion

The conversion pipeline has three stages plus downstream retraining. **(1) Native-speaker audit:** a native Indian SMS user (Hindi, Hinglish, and English-India literate) reviews 100 stratified positives and flags 18 as non-fraud rather than fraud. These prefix-based patterns, largely commercial marketing but also including legitimate institutional notices, cover 31 near-duplicate rows and are stored in the historical “marketing” bin. For 19, both judges also return marketing; for 3, both return fraud; the remaining 9 are mixed or unclear, including 8 with one fraud verdict. The native-speaker decision overrides the judges for all 31 rows. **(2) Dual-LLM consensus:** each pre-dedup source-positive candidate is judged independently by GPT-4o and Gem-

ini 2.5 Flash, which provide strong coverage across Hindi, Hinglish, and English-India; the resulting judge-baseline coupling for those two models is treated as a limitation (§6). Claude Haiku 4.5 is held out as a non-judge closed-source comparator. The complete adjudication prompt and the distinct minimal inference prompt are reproduced in Appendix A. **(3) Consensus rule:** a message keeps its positive label if both judges agree on fraud; flips to safe if both agree on marketing; is dropped if the judges disagree or either says unclear. The 18 manual non-fraud flags override. Table 1 reports 7,234 source-positive rows after source deduplication. Inter-judge agreement is 93.5%. Three annotators label 276 stratified messages over all four categories at Fleiss’ $\kappa = 0.821$ and Krippendorff’s $\alpha = 0.821$; 127 rows have an unclear or tied majority, so agreement with the LLM policy is reported only for the remaining 149 (§6). **Retraining:** the policy-aligned label set retrains XLM-RoBERTa-base from scratch with identical hyperparameters as the noisy-label baseline. The consensus-rule architecture is adaptable to other binary corpora with suspected label noise, though prompts and gold sets are domain-specific.

3.3 Indian institutional SMS template taxonomy

A core observation under the original minimal-prompt condition is that LLM API misclassification is concentrated on a small number of repeated legitimate institutional templates, not on conversational content. To measure this, we define a deterministic regex taxonomy with five primary institutional families (**OTP**, **Aadhaar/KYC**, **UPI/Payment**, **Bank/account**, **EMI/loan**) plus three secondary buckets (Reward/lottery, Marketing/promo, URL/click) plus an “Other” residual. Each message is assigned to at most one bucket via first-match-wins ordering, most specific to most generic. We use one published deterministic regex taxonomy for all reported systems; the same taxonomy is applied uniformly across the eight original systems and is released in the artifact. The five institutional families together account for 319 (24.7%) of the 1,294 policy-negative (label=0) messages in the cleaned test set, but absorb 49–58% of the LLM zero-shot false-positive volume across the four closed-source LLMs—i.e., institutional templates are roughly $2\times$ over-represented in commercial-LLM false-alarm volume relative to their share of legitimate messages. Our lo-

cally fine-tuned model generates zero false positives from the three multi-row institutional families (UPI/Payment, Bank/account, EMI/loan) on the cleaned test set.

3.4 Model, training, and LLM-baseline protocol

Encoder. We fine-tune FacebookAI/xlm-roberta-base (270M parameters) (Conneau et al., 2020), a multilingual successor in the BERT (Devlin et al., 2019)/DistilBERT (Sanh et al., 2019) family, on the cleaned multilingual training split (Indian SMS plus 30K English samples drawn at random from a global scam-detection corpus for cross-lingual coverage), 80/10/10 stratified by (language, label) with seed=42. Optimizer: AdamW, lr= 2×10^{-5} , weight decay=0.01, batch size=16, three epochs, class-weighted cross-entropy. Training takes 50 minutes on a single Apple M2 Max GPU.

LLM baselines. We evaluate four closed-source LLMs (GPT-4o, GPT-4o-mini, Claude Haiku 4.5, Gemini 2.5 Flash) plus two open-weight LLMs via Together AI (Llama-3.3-70B-Instruct, Qwen-2.5-7B-Instruct). All six receive the identical English minimal system prompt reproduced in Appendix A. We use the original Hindi/Hinglish/English-India text, temperature 0.0, and a single-word parser.

Prompt-sensitivity protocol. After observing the zero-shot failure, we rerun the four closed-source APIs with a same-session zero-shot control and a ladder comprising policy definitions (P1), one legitimate bank example, seven hand-written safe/fraud examples (P2), and P2 plus an explicit institutional-message carve-out (P3). P2 is the main strong-prompt condition. Its examples are not copied from the test set (maximum token-Jaccard overlap 0.27), but they were designed with knowledge of the failure mode; these are exploratory sensitivity results, not a preregistered held-out prompt search. The strict parser accepts only an exact `scam` or `safe` string after whitespace trimming; all other outputs are unparseable. Metrics exclude those rows and report their count. Complete prompts and parsed per-message decisions are released; verbatim responses are withheld because a few echoed input fragments.

Statistics. We report F1 (binary positive class), Matthews correlation coefficient (MCC), balanced accuracy, false-positive rate (FPR), precision, and recall, each with 95% bootstrap confidence intervals (1,000 resamples, seed=42). Pairwise sig-

Table 2: Original minimal-prompt comparison on the policy-aligned test set ($n = 1,393$; 99 positives). FPR is on 1,294 policy-negative rows and is the primary endpoint. F1/MCC/recall are policy-dependent secondary measures.

Model	F1	MCC	FPR	Rec
XLM-R cleaned (ours)	.965	.963	.004	.980
Gemini 2.5 Flash	.558	.583	.121	1.00
GPT-4o	.512	.542	.146	1.00
Claude Haiku 4.5	.441	.476	.191	.990
XLM-R EN-only	.401	.433	.220	.970
Llama-3.3-70B	.396	.435	.233	1.00
GPT-4o-mini	.310	.348	.340	1.00
Qwen-2.5-7B	.285	.320	.383	1.00

nificance against our model uses McNemar’s test (McNemar, 1947) with continuity correction on the same 1,393 row-aligned test set, with Holm-Bonferroni multiplicity correction (Holm, 1979) across the seven original zero-shot comparisons. Because two positive-label judges are also baselines, F1, MCC, and recall for GPT-4o and Gemini are judge-shaped secondary measures. FPR on the 1,294 policy-negative rows is the primary endpoint; we separately report the 1,222-row untouched-source-negative check. For XLM-R probabilities we additionally report 10-bin equal-width expected calibration error (ECE) and Brier score.

4 Results

4.1 Default zero-shot comparison

Table 2 reproduces the original comparison on the 1,393-message test (99 policy-positive, 1,294 policy-negative). Its primary column is FPR: the local XLM-R records 0.4%, while the six minimal-prompt APIs range from 12.1% to 38.3%. At the test mixture, 12.1% FPR corresponds to roughly 121,000 false positives per million legitimate messages; this is a normalization, not a forecast for an operator’s traffic. XLM-R obtains F1=0.965 [0.937, 0.989] and MCC=0.963. These positive-class metrics depend on our policy labels, and GPT-4o/Gemini have the additional judge-baseline coupling; they are included for context rather than as independent evidence.

Under this prompt, every API has recall ≥ 0.99 but low precision, a signature of positive over-prediction at 7.1% prevalence. The English-only XLM-R control (F1=0.401, FPR=22.0%) shows that an encoder without policy-aligned Indian training data does not itself solve the task. These observations apply to this prompt and snapshot; §4.2

shows that stronger instructions substantially alter the result.

Restricting FPR to the 1,222 untouched source-negative rows gives 0.08% for the local model, versus 10.88–35.19% for the six minimal-prompt APIs and 18.66% for English-only XLM-R. Thus the false-positive gap does not depend on the 72 source-positive rows converted to safe.

Judges-excluded sensitivity. Removing GPT-4o and Gemini from the baseline rows leaves the local model ahead of the remaining comparators (Appendix F). This checks direct judge–baseline reuse, but it does not make the positive target human-defined: XLM-R and all remaining F1 values still depend on the same policy conversion. We therefore do not use this ranking as the headline evidence.

Paired significance. For the original comparison, all seven paired McNemar tests survive Holm correction at $p < 10^{-28}$ (full counts in Appendix C). Statistical separation does not resolve target validity, hence the emphasis on untouched-negative FPR and sensitivity audits.

4.2 Strong-prompt and consensus sensitivity

Table 3 reports P2, which supplies the fraud policy and seven examples, plus GPT-4o–Gemini consensus. These exploratory prompts were created after error inspection. Compared with same-session zero-shot reruns, P2 reduces global FPR from 14.8% to 1.5% for GPT-4o and from 19.6% to 1.8% for Claude; both reach approximately zero Bank/UIP FPR. The original conclusion that all LLM APIs over-flag these templates is therefore not stable to prompting.

GPT-4o P2 misses both the 1% global-FPR gate and, narrowly, the 0.95 policy-label recall floor; its more restrictive P3 prompt meets the FPR gates at 0.9% global FPR but lowers policy-label recall to 0.909. Claude P2/P3 remains above 1% global FPR. Gemini and GPT-4o-mini trade recall or global FPR for template restraint. The AND consensus meets the FPR gate but policy-label recall falls to 0.758; OR recovers recall at 3.6% FPR. Because the positive target partly uses GPT-4o/Gemini judgments, these recall values are secondary. This is a deployment trade-off result, not evidence of inherent LLM incapability or an overall model ranking.

4.3 Template-level false-positive analysis

Table 4 reports per-template FPR for the original minimal-prompt comparison under one re-

leased regex partition. If each positive verdict were mapped directly to blocking, GPT-4o-mini would flag 77.3% of the 163 Bank/account and 56.9% of the 146 UPI/Payment policy-negative messages; the local model flags none. These are benchmark outcomes, not estimates from a bank’s production distribution.

All six APIs over-predict fraud on both Bank and UPI under the minimal prompt, although magnitude varies substantially by model. The local model’s 0.0% rate shows that supervised exposure can fit these benchmark templates; it does not establish performance on unseen institutions. Figure 1 visualizes only this original zero-shot condition.

Negative-label audit. We subsequently asked both label judges to classify all 1,294 policy-negative rows: 1,111 (85.9%) were called safe by both, 141 received mixed verdicts, and 42 (3.25%) were called fraud by both. Among the 1,222 untouched source negatives, the split is 1,065 safe, 117 mixed, and 40 fraud; 32 of those 40 are synthetic UPI messages. Among the 72 converted source positives, it is 46/24/2. This is neither re-labeling nor independent gold; it exposes judge behavior and motivates reporting the untouched-source-negative check separately.

Human-label check. Three annotators label the same 276 rows using *fraud/marketing/legitimate-institution/unclear*. Fleiss’ $\kappa = 0.821$ and Krippendorff’s $\alpha = 0.821$ include all four categories. Annotator A assigns 2/62/73/139 rows to those categories, versus 40/38/72/126 for B and 46/39/68/123 for C. Only 149 rows have a resolved non-unclear majority; 131/149 (87.9%) match the binary LLM policy, while 127 are unclear or tied. Under a binary human-majority mapping (fraud versus all non-fraud categories), 33 Bank and 8 UPI rows are majority-safe. XLM-R false-flags 1/33 (3.0%) and 0/8; GPT-4o-mini false-flags 21/33 (63.6%) and 7/8 (87.5%). These are majority, not unanimous, labels and remain small post-hoc subsets.

Structural-duplicate sensitivity. Masking digits, accounts, amounts, and URLs creates 1,219 structural clusters from 1,393 rows. Keeping one row per cluster leaves local Bank/UIP FPR at 0.0% ($n = 104/124$); in same-session zero-shot predictions, GPT-4o-mini remains at 61.5%/52.4%. Other API rates move in both directions. This reduces repeated-template weighting but is not a train/test split by institution, which remains future work.

The English-only XLM-R control shows that

Table 3: Strong-prompt P2 and consensus sensitivity on the same test. Metrics exclude unparseable responses (Unp). Consensus uses GPT-4o and Gemini P2; one Gemini output is unparseable. F1 and recall remain policy-dependent.

System	Bank FPR	UPI FPR	Global FPR	Recall	F1	Unp
XLM-R local	0.0%	0.0%	0.4%	.980	.965	0
GPT-4o P2	0.0%	0.0%	1.5%	.949	.887	0
Claude P2	0.0%	0.7%	1.8%	.960	.876	40
Gemini P2	0.0%	3.4%	2.7%	.778	.730	1
GPT-4o-mini P2	0.0%	0.0%	15.8%	.990	.488	0
GPT-4o+Gemini AND	0.0%	0.0%	0.6%	.758	.824	1
GPT-4o+Gemini OR	0.0%	3.4%	3.6%	.970	.797	1

Table 4: Minimal-prompt false-positive rate (%) on policy-negative messages within each Indian institutional SMS template, across the eight original systems. Buckets with $n < 10$ (EMI/loan, Marketing, OTP, KYC, Reward) are directional only; we headline UPI/Payment, Bank/account, URL/click, and Other. “Ours” = XLM-R cleaned; “EN-only” = XLM-R English-only. Two Bank rows and no UPI rows are converted source positives; the remainder in those two buckets retain source-negative labels.

Template	n	Ours	mini	4o	Claude	Gemini	Llama	Qwen	EN-only
UPI/Payment	146	0.0	56.9	31.5	39.7	27.4	25.3	5.5	48.6
Bank/account	163	0.0	77.3	28.8	49.1	24.5	74.2	65.6	57.1
EMI/loan	8	0.0	62.5	25.0	37.5	12.5	50.0	100.0	87.5
Marketing	6	0.0	83.3	50.0	66.7	50.0	83.3	83.3	83.3
URL/click	21	0.0	4.8	4.8	47.6	4.8	4.8	71.4	9.5
Other	946	0.5	23.2	9.5	9.6	7.6	14.1	37.2	11.0

	UPI/Payment FPR $\leq 1\%$	Bank/account FPR $\leq 1\%$	EMI/loan FPR $\leq 1\%$	Global FPR $\leq 1\%$	Recall ≥ 0.95
XLM-R cleaned (ours)	✓ 0.0%	✓ 0.0%	✓ 0.0%	✓ 0.4%	✓ 0.980
Gemini 2.5 Flash	✗ 27.4%	✗ 24.5%	✗ 12.5%	✗ 12.1%	✓ 1.000
GPT-4o	✗ 31.5%	✗ 28.8%	✗ 25.0%	✗ 14.6%	✓ 1.000
Claude Haiku 4.5	✗ 39.7%	✗ 49.1%	✗ 37.5%	✗ 19.1%	✓ 0.990
XLM-R English-only	✗ 48.6%	✗ 57.1%	✗ 87.5%	✗ 22.0%	✓ 0.970
Llama-3.3-70B	✗ 25.3%	✗ 74.2%	✗ 50.0%	✗ 23.3%	✓ 1.000
GPT-4o-mini	✗ 56.9%	✗ 77.3%	✗ 62.5%	✗ 34.0%	✓ 1.000
Qwen-2.5-7B	✗ 5.5%	✗ 65.6%	✗ 100.0%	✗ 38.3%	✓ 1.000

Figure 1: Pass/fail matrix for the original minimal-prompt comparison: eight systems against five illustrative gates (per-template FPR $\leq 1\%$ on UPI/Payment, Bank/account, EMI/loan; global FPR $\leq 1\%$; recall ≥ 0.95). Cells colored green for pass, red for fail. XLM-R passes every gate; the seven original comparators fail four of five under this condition (only recall passes). EMI/loan $n = 8$ is directional.

this is not encoder architecture alone: fine-tuned on English-only scam data, it false-flags 57.1% of Bank/account and 48.6% of UPI/Payment SMS. Policy-aligned Indian SMS training exposure is the measured difference between the two encoder controls.

Per-language F1. Our cleaned XLM-R scores 0.97 on Devanagari Hindi ($n = 471$), 0.74 on Hinglish ($n = 169$), 1.00 on English-India ($n = 746$); Gemini scores 0.63/0.33/0.56 and English-only XLM-R 0.62/0.21/0.37. The biggest lift is on English-India (+0.63 over English-only), so the gap is content familiarity, not script. Hinglish is smallest in our test and noisier (iiitd’s positives shrank $17\times$ under cleaning).

4.4 Operating-point analysis

Table 5 sweeps four decision thresholds for XLM-R. UPI/Payment and Bank/account FPR stay at 0.0% at every threshold; recall stays ≥ 0.929 down to $\tau = 0.90$ and global FPR drops from 0.54% to 0.15%. In deployment, score thresholds can map to *allow*, *warn*, *manual review*, or *block* actions rather than a single binary block decision. The full seven-threshold sweep is in Appendix D; a label-policy sensitivity analysis (Appendix E) reports performance under four target-label variants. At $\tau = 0.50$, the model’s probability scores have 10-bin ECE=0.0066 and Brier score=0.0032 on this test set; these in-sample-distribution calibration values do not guarantee calibration after deployment shift.

Table 5: Operating-point sweep for the cleaned XLM-R on the 1,393-row test. Flag/M = messages flagged per 1M-SMS/day, projected at the held-out test set prevalence and template mix; operators should recompute on their own traffic.

τ	Rec	Glob	UPI	Bank	Flag/M
0.30	1.000	0.54%	0.0%	0.0%	76,095
0.50	0.980	0.39%	0.0%	0.0%	73,223
0.80	0.960	0.15%	0.0%	0.0%	69,634
0.90	0.929	0.15%	0.0%	0.0%	67,480

5 Lessons Learned

(1) Label policy is a deployment control, not a cleaning detail. Under our fraud-blocking definition, 77.5% of iitd’s “spam” positives map to marketing. This is policy conversion, not correction of the source task.

(2) Template-level FPR exposes concentrated error. GPT-4o-mini’s 34.0% global FPR in the original minimal-prompt run includes 77.3% Bank/account and 56.9% UPI/Payment FPR; an aggregate score obscures this concentration.

(3) Prompt choice is part of the audited system. Policy definitions and examples remove the headline Bank/UI errors for GPT-4o, but other global FPR or recall failures remain. A “model” result without its prompt and parser is not a reproducible deployment result.

(4) Local supervision is a strong benchmark option, not production proof. On this split, XLM-R meets the FPR and policy-label recall gates and avoids external API transfer. Independent traffic, unseen institutions, adaptive attacks, and operational monitoring remain required before deployment.

6 Conclusion

On this benchmark, six default zero-shot APIs over-flag Indian institutional SMS, while policy-grounded prompts greatly reduce the error for some models. GPT-4o P3 and the AND consensus meet the illustrative FPR gates with lower policy-label recall; locally supervised XLM-R meets both FPR and recall gates on this split. Since the positive target is judge-shaped, this is not an overall ranking. The durable contribution is the evaluation pattern: define the fraud policy, preserve judge-independent endpoints, disclose prompt sensitivity, and gate deployment by template-specific harm. Independent human-labeled and unseen-institution evaluation is still needed before production use.

Limitations

Construct and label validity. Generic spam, financial fraud, and a deployment block decision are different constructs. Our conversion is one operational fraud policy, not correction of the source datasets. Only source positives were candidates for conversion. Consequently, the policy-negative test class contains 1,222 untouched source negatives and 72 converted source positives. The post-hoc negative-judge audit measures LLM agreement, not truth. The 30K global-English training augmentation retains its source labels and was not subjected to either the Indian positive-class conversion or a separate negative-label audit. Alternative policies change F1 (Appendix E).

Judge coupling and limited human validity. GPT-4o and Gemini helped construct the positive labels and are also evaluated. Removing their baseline rows (Appendix F) does not remove judge influence from the positive target or the XLM-R training labels. Accordingly, recall and F1 are secondary for all systems and especially coupled for those two APIs. We did not rebuild labels or retrain under alternative dual-judge consensus or manual-override rules; Appendix E only perturbs frozen target labels post hoc. A fully human-labeled independent evaluation set is not available.

The 276-row human study includes 127 unclear or tied majorities, and its 33 Bank plus 8 UPI majority-safe rows are too small to settle validity. Detailed annotator recruitment, independence, and compensation are not available in the released record, further limiting interpretation.

Synthetic and repeated templates. The UPI source-negative subset is synthetic, including repeated TATA Power and transfer-receipt templates. It tests restraint on designed legitimate messages, not observed bank traffic. The original fingerprint deduplication leaves structural near-duplicates. One-per-test-cluster sensitivity reduces their weight but does not remove train–test template overlap. We did not rebuild the released split after structural deduplication or perform leave-cluster-out retraining. We have no institution-held-out, temporal, or external production test, so the local model’s 0% Bank/UI FPR may reflect in-market template exposure.

Prompt and model snapshot. The P1–P3 prompts were designed after seeing zero-shot errors and therefore estimate attainable prompt sensitivity, not unbiased prompt-selection performance.

Unparseable outputs are excluded and explicitly counted. API aliases and hosted model behavior can change; results describe the recorded May–July 2026 calls and released outputs, not future versions. We also do not evaluate retrieval augmentation: retrieval cannot by itself define an operator’s action policy, though it may help on unseen threats and remains useful future work.

Sample size and release gates. The test has 99 positives. Several policy-negative buckets are small (EMI $n = 8$, OTP $n = 1$, KYC $n = 1$, Reward $n = 2$), so only Bank ($n = 163$) and UPI ($n = 146$) support the institutional headline. The 1% FPR and 0.95 recall gates illustrate a conservative audit; they are neither statistically derived operating optima nor regulations.

Unfinished external-validity checks. We did not add observed, non-synthetic UPI negatives, validate the regex taxonomy assignments with a separate human study, or construct the substantially larger fully human-gold test requested during review. The taxonomy is a deterministic analysis layer, not a human-validated ontology. These require licensed representative traffic, fresh annotation, and a pre-specified evaluation and are future work rather than camera-ready claims.

Deployment and adversaries. Latency was measured on one Apple M2 Max and will differ on other hardware. We do not evaluate adaptive evasion, network metadata, sender reputation, URL resolution, throughput under load, human review operations, or post-deployment drift. Adaptive attacks often break static defenses (Zhan et al., 2025); these checks, external validation, and policy review are required before any production claim.

Ethical Considerations

Data. Training data comes from publicly described or research-accessible datasets. CloveAI declares MIT and UCI declares CC-BY-4.0; some other source repositories do not state an explicit dataset license, and the IIITD distribution is available on request without an explicit redistribution license in its README. The archival artifact uses normalized SHA-256 values for IIITD rows rather than redistributing their message text; the published archive does not include an automated IIITD reconstruction script. Some corpora may contain residual PII despite redaction by the dataset creators; the released checkpoint is a binary classifier whose output is a single probability and not a generative

model, mitigating extraction risk. We have not run formal PII extraction probes against the checkpoint and treat that as future work.

Dual use. Our findings could inform an adversary or operator aware of which institutional templates LLM APIs systematically misclassify. The released examples are redacted and the taxonomy describes broad message families rather than evasion instructions. We judge the value of reproducible defensive auditing to outweigh this limited disclosure risk.

External API processing. The benchmark messages sent to commercial APIs come from public research datasets rather than private customer traffic. Production use could transfer sensitive financial communications to external providers and would require provider, retention, jurisdiction, and consent review; this paper performs no such compliance assessment.

False positives in deployment. The operational implication of our minimal-prompt results is that mapping every positive verdict directly to a block could suppress legitimate financial notifications. Strong prompting mitigates this for some APIs, reinforcing that no tested output should be silently converted into production blocking without local validation and an appeal or review path.

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A Exact Adjudication and Minimal Inference Prompts

The positive-class adjudication and model inference calls serve different tasks and therefore use different prompts. The complete adjudication system prompt is:

You are classifying an Indian SMS message that another dataset labeled as “spam”. Decide whether the message is FRAUD (financial scam, phishing, impersonation, deceptive intent, Aadhaar/UPI/KYC manipulation, fake prize claims, premium-rate dial-in traps, identity theft, predatory loan with deceptive terms) or MARKETING (legitimate aggressive commercial promotion from a real or plausible Indian company: retail, telecom plans, real estate, health checkups, taxi services, etc., even if annoying). If you cannot decide, reply UNCLEAR.

Reply with ONLY ONE WORD: fraud, marketing, or unclear.

The complete minimal inference system prompt used for the original LLM comparison is:

You are a scam detection classifier. Given a message in any language, respond with ONLY a single word: “scam” if the message is a scam/phishing/fraud attempt, or “safe” if it is legitimate. No explanation, just one word.

B Full Prompt-Ladder Results

P1 adds a fraud-policy definition to the baseline. One-shot adds one safe bank-transfer example. P2 adds seven hand-written examples: safe bank credit, OTP, retail promotion, and EMI reminder; and fraudulent KYC link, fee-based prize, and credential-theft impersonation. P3 adds the explicit instruction that bank, UPI, account, amount, and OTP vocabulary alone is not evidence of fraud. All conditions request exactly `scam` or `safe`. The released script contains the complete prompt strings and contamination check.

C Paired McNemar Detail

We report the full discordance detail behind the McNemar comparisons referenced in §4. For each baseline, b counts the rows our cleaned XLM-R got wrong while the baseline got right and c counts the reverse. The c/b ratio shows how lopsided each comparison is in favour of our model. All seven adjusted p -values are below 10^{-28} after Holm-Bonferroni correction.

D Full Operating-Point Sweep

Table 8 reports the full seven-threshold operating-point sweep for the cleaned XLM-R on the 1,393-row test set. EMI/loan ($n_{\text{safe}} = 8$) and Marketing/promo ($n_{\text{safe}} = 6$) FPRs are directional only because of small denominators; we include them here for completeness. Across all seven thresholds, per-template FPR on UPI/Payment, Bank/account, EMI/loan, and Marketing/promo is 0.0%.

E Label-Policy Sensitivity

We re-evaluate all eight original systems under four target policies: (A) the body policy; and (B1, B2, B3) the same 1,393 rows after reclassifying current policy-positive rows from *cloveai*, *iiitd*, or *vinit9638* as safe. No rows are removed and no model is retrained, so these are target-label perturbations, not source-holdout generalization tests. XLM-R has the highest F1 under each, but the mechanically derived variants do not establish construct validity.

F Judges-Excluded Sensitivity

We remove the two label-construction judges (GPT-4o and Gemini) from the baseline rows. Table 10 reports the remaining systems on the same 1,393-row test. The local model leads the other rows by

F1 margins of 0.524–0.680 and FPR margins of 18.7–37.9 percentage points. This removes direct reuse for the compared API rows, but the local model was trained on the policy-converted labels and every positive-class metric still depends on that target.

Paired McNemar tests against each remaining baseline are below 10^{-28} after Holm correction (Table 7). The cleaner evidence is the separately reported 1,222-row untouched-source-negative FPR check; Table 4 otherwise uses all 1,294 policy-negative rows.

G Redacted Examples

Table 11 shows redacted examples from the cleaned test set organized into four named failure-case categories. Long numerals (account numbers, OTPs, transaction IDs) are masked to [NUM]; currency amounts to [AMT]; URLs to [URL]; the obfuscated digit-suffix on A/C masks to XX[ACC]. Personal names left unmasked are taken from the original synthetic *en_in* pool (*cloveai_india*), which is itself synthetic. LLM column shows GPT-4o / GPT-4o-mini / Claude / Gemini outputs (1=scam, 0=safe). ISO language codes: *hi* = Devanagari Hindi, *hi_en* = Hinglish, *en_in* = English-India.

Category 1 – Policy-positive fraud caught by our model. Messages labeled fraud under the study policy that XLM-R and the original zero-shot closed-source LLMs flag (rows 1–3 in Table 11). These span aggressive promotional-bait fraud (row 1, fake utility-bill discount with suspicious URL), prize-bait fraud (row 2, “free gift pack” with handling-fee hook), and impersonation fraud (row 3, fake Instagram account-suspension notice).

Category 2 – Zero-shot false positive on UPI/Payment. Source-negative synthetic UPI notifications that original zero-shot APIs flag but XLM-R passes (rows 4–6). They imitate billing prompts and transfer receipts; they are not sampled bank traffic. Directly mapping these verdicts to blocks would suppress the benchmark messages.

Category 3 – Zero-shot false positive on Bank/account. Source-negative balance updates and card-spend confirmations that original zero-shot APIs flag but XLM-R passes (rows 7–8). GPT-4o-mini flags 77.3% of this 163-row subset (Table 4).

Table 6: Full prompt ladder. Zero is a same-session rerun of the original minimal prompt. Metrics exclude unparseable responses (Unp). The search was designed after observing errors and is exploratory.

Model	Condition	Bank FPR	UPI FPR	Global FPR	Recall	F1	Unp
Claude	Zero	50.3%	39.7%	19.6%	.990	.436	0
	P1	0.0%	1.4%	3.6%	1.00	.811	9
	One-shot	3.1%	1.4%	3.6%	.980	.805	28
	P2	0.0%	0.7%	1.8%	.960	.876	40
	P3	0.0%	0.7%	1.7%	.960	.880	18
Gemini	Zero	24.5%	27.4%	12.3%	1.00	.555	1
	P1	1.8%	28.1%	7.9%	.879	.604	0
	One-shot	0.0%	13.7%	7.1%	.848	.611	0
	P2	0.0%	3.4%	2.7%	.778	.730	1
	P3	0.0%	1.4%	1.2%	.737	.781	0
GPT-4o	Zero	31.3%	30.8%	14.8%	1.00	.509	0
	P1	0.0%	0.0%	7.1%	1.00	.683	0
	One-shot	0.0%	0.0%	4.9%	.990	.754	0
	P2	0.0%	0.0%	1.5%	.949	.887	0
	P3	0.0%	0.0%	0.9%	.909	.900	0
GPT-4o-mini	Zero	75.5%	59.6%	34.2%	1.00	.309	0
	P1	0.0%	19.2%	25.0%	1.00	.380	0
	One-shot	1.2%	15.8%	37.9%	1.00	.288	0
	P2	0.0%	0.0%	15.8%	.990	.488	0
	P3	1.2%	0.0%	23.7%	1.00	.392	0

Table 7: Paired McNemar tests (continuity correction) vs. our cleaned XLM-R on the 1,393-sample aligned test. b = our model wrong, other correct; c = our model correct, other wrong. Holm-Bonferroni adjusts across 7 comparisons; all adjusted p -values are below 10^{-28} .

Comparison vs. ours	b	c	c/b
Qwen-2.5-7B	3	492	164×
GPT-4o-mini	4	437	109×
Llama-3.3-70B	2	297	149×
XLM-R English-only	3	283	94×
Claude Haiku 4.5	4	245	61×
GPT-4o	4	186	47×
Gemini 2.5 Flash	4	154	39×

Category 4 – Our model’s residual errors. The five residual false positives + two residual false negatives left by our cleaned XLM-R. Row 9 (false positive) is a short interactive quiz-style message whose surface form overlaps with fraud bait; the operating-point *warn/manual review* bands (§4.4) route exactly this kind of case for human decision rather than block. Row 10 (false negative) is a Forbes India “subscribe and win” promotion that all four closed-source LLMs flag (correctly, in their favor) but our model marginally clears at $\tau = 0.50$ ($p_{\text{scam}} = 0.470$).

H Dataset Sources, Licenses, and Privacy Statement

Privacy and redistribution statement. The compact camera-ready supplement contains no raw SMS text or verbatim provider responses; it releases parsed per-message decisions plus sanitized

Table 8: Full seven-threshold operating-point sweep for the cleaned XLM-R. Per-template FPR on UPI/Payment, Bank/account, EMI/loan, and Marketing/promo is 0.0% at every threshold. “Flag/M” projects messages flagged per 1M-SMS/day at the held-out test set prevalence and template mix; operators should recompute on their own traffic.

τ	Recall	Glob FPR	Other FPR	Flag/M
0.30	1.000	0.54%	0.63%	76,095
0.40	1.000	0.39%	0.53%	74,659
0.50	0.980	0.39%	0.53%	73,223
0.60	0.980	0.23%	0.32%	71,788
0.70	0.980	0.23%	0.32%	71,788
0.80	0.960	0.15%	0.21%	69,634
0.90	0.929	0.15%	0.21%	67,480

response status. The archival artifact releases normalized SHA-256 values, cleaned labels, and source metadata for IIITD rather than its message text; users must obtain the official distribution separately, and the current archive does not provide automated reconstruction. We cite the original IIITD paper (Yadav et al., 2011) and reproduce the project contact (Prof. Ponnurangam Kumaraguru, pk@iiitd.ac.in). We collected no private customer traffic. Source corpora may contain residual identifiers; we do not claim that every long digit sequence is masked or synthetic. Personal names in synthetic English-India templates from *cloveai_india* are themselves synthetic (e.g. “Re-fund INR Oak-Sridhar initiated”) and do not correspond to identifiable individuals.

Artifact availability. The frozen supplementary artifact contains the cleaned 1,393-row test (with 169 controlled-source IIITD texts represented only

Table 9: Regenerated label-policy sensitivity (F1) across eight original systems. Parentheses give the positive denominator. “Ours” = policy-aligned XLM-R; “Eng-only” = XLM-R English-only.

Policy	Ours	mini	4o	Claude	Gemini	Llama	Qwen	EN-only
A current (99)	.965	.310	.512	.441	.558	.396	.285	.401
B1 cloveai + $\rightarrow 0$ (45)	.585	.154	.270	.226	.299	.202	.141	.198
B2 iitd + $\rightarrow 0$ (91)	.933	.289	.480	.413	.524	.370	.265	.374
B3 vinit + $\rightarrow 0$ (64)	.759	.212	.364	.313	.400	.275	.194	.288

Table 10: Judges-excluded sensitivity. GPT-4o and Gemini 2.5 Flash are removed from baseline rows because they participated in positive-label policy construction (§3.2). Remaining systems use the same 1,393-row test. Sorted by F1.

Model	F1	MCC	FPR
XLM-R cleaned (ours)	0.965	0.963	0.004
Claude Haiku 4.5	0.441	0.476	0.191
XLM-R English-only	0.401	0.433	0.220
Llama-3.3-70B (open)	0.396	0.435	0.233
GPT-4o-mini	0.310	0.348	0.340
Qwen-2.5-7B (open)	0.285	0.320	0.383

by hashes), a 7,234-row audit, both XLM-R checkpoints, frozen predictions/probabilities, historical training/evaluation source snapshots, the regex taxonomy, a three-annotator package, and environment files. Required training modules and the IIITD reconstruction path are absent, so the archive does not reproduce training from scratch or full checkpoint re-inference as packaged; the controlled IIITD source must be obtained and rejoined locally. It is archived on Zenodo at <https://doi.org/10.5281/zenodo.20480151> (approximately 1.66 GB, Open Access). Apache-2.0 covers the project code/documentation; included data retain their upstream terms. The compact GroundLM camera-ready supplement is archived separately at <https://doi.org/10.5281/zenodo.22732001> (version v5-groundlm, approximately 165 kB). It adds prompt-ladder predictions, regenerated label-policy sensitivity, consensus, negative-label, annotation, structural-cluster, and calibration results. Its description corrects earlier artifact claims. Both records should be consulted together; the reproduction limitations above remain. Aggregate-only analyses cannot be independently reconstructed from the compact supplement.

I Pre-Deployment Audit Checklist

A compact operator-facing checklist distills the paper’s audit framing. The thresholds are illustrative;

operators should derive their own from traffic, harm costs, legal review, and review-team capacity.

1. **Target policy written before scoring.** Distinguish source spam, study fraud, and operational block/review/allow actions; document how marketing and unclear messages are handled.
2. **Gold evidence is independent enough for the claim.** Preserve untouched-source-negative and human-labeled endpoints; disclose any model used both to construct labels and to predict them.
3. **Per-template and global FPR meet local budgets.** Include high-volume institutional templates and confidence intervals; do not infer production rates from a synthetic or repeated-template mixture.
4. **Recall and calibration meet the chosen operating point.** Report the FPR–recall trade-off, probability calibration, and the positive label definition on which recall depends.
5. **Prompt, parser, model version, and failure behavior are pinned.** Re-test after any provider or prompt change, and treat unparseable or API-error handling as part of the system.
6. **Uncertain cases have a review and appeal path.** Project review volume on the operator’s traffic and avoid silently mapping every flag to a block.
7. **Independent deployment checks are complete.** Evaluate unseen institutions, temporal shift, adaptive attacks, latency under load, cost, privacy, retention, and applicable policy before rollout.

Passing the numerical gates on this benchmark is necessary only for this test scenario, not sufficient for deployment. The checklist deliberately extends beyond what this paper empirically validates.

Table 11: Redacted examples from the 1,393-row test. “True” is the study policy label (1=fraud, 0=safe). “Ours” is XLM-R; “4LLMs” reports the original minimal-prompt GPT-4o/GPT-4o-mini/Claude/Gemini outputs.

#	Cat.	Lang	Template	Message (redacted)	True	Ours	4LLMs
1	1	en_in	Mktg	Get 50% discount on your [NUM] electricity bill! Pay only INR [AMT]. Limited time: [URL]	1	1	1/1/1/1
2	1	hi_en	Reward	New year FREE gift pack of 25 Aroma candles n Holders MRP Rs [AMT] fr members just pay Rs [AMT] as handling charges	1	1	1/1/1/1
3	1	en_in	Other	Your Instagram account is suspended. Appeal now!	1	1	1/1/1/1
4	2	en_in	UPI/Pay	TATA Power: Bill for August is INR [AMT]. Due: [NUM]-09-04. Pay via UPI: [NUM]@ok[NUM]	0	0	1/1/1/1
5	2	en_in	UPI/Pay	UPI of INR [AMT] to [NUM] from A/C XX[ACC] successful. Ref No. Kumer – Yes Bank	0	0	1/1/0/0
6	2	en_in	UPI/Pay	UPI of INR [AMT] to [NUM] from A/C XX[ACC] successful. Ref No. Dutt – Yes Bank	0	0	0/1/0/0
7	3	en_in	Bank/acc	Your PNB account XX[ACC] has balance of INR [AMT] as on [NUM].	0	0	0/1/1/0
8	3	en_in	Bank/acc	Thank you for using your ICICI Bank Debit Card XX[ACC] for INR [AMT] at [NUM]. Avl Bal INR Arya.	0	0	1/1/1/0
9	4	hi_en	Other	The Final Hurrah! Who won the final race of the [NUM] F1 season? 1) Mark Webber 2) Sebastian Vettel 3) Lewis Hamilton	0	1	1/1/1/1
10	4	hi_en	Other	Win a Mercedes-Benz & assured gifts by subscribing to the Forbes India special offer. SMS Forbes to [NUM] NOW. T&C	1	0	1/1/1/1

Table 12: Public Indian and English SMS sources used for training and held-out evaluation. “Raw” is the source-positive count; “Use” describes the policy-aligned Indian mix. English augmentation retains source labels. Sources are publicly described or research-accessible; no proprietary or operator-internal SMS data is used.

Source	Language	Raw	Use	License	Provenance / Citation
cloveai_india	English-India	5,558	Train+test	MIT	HuggingFace dataset
vinit9638	Devanagari	626	Train+test	Not stated	GitHub repository
vinit9638_multi	Hindi	625	Dedup’d	Not stated	Same GitHub source , Hindi-language filter
iiitd	Devanagari	995	Train+test (hash only in artifact)	Avail. on req.; no explicit redistribution licence	IIIT-Delhi SMSAssassin (Yadav et al., 2011)
flatmars	Hinglish (hi_en)	55	Train+test	Not stated	GitHub repository
English augmentation (source labels; XLM-R train only, not in the 1,393-row test):					
ealvaradob	Mixed	—	Train+test	Not stated	GitHub repository
UCI SMS Spam	English	—	Train only	Apache-2.0	HuggingFace dataset
SpamAssassin	English	—	Train only	CC-BY-4.0	UCI ML Repository
Enron	English	—	Train only	Source terms	spamassassin.apache.org
				Source terms	CMU Enron corpus